

Measuring AI Ability to Complete Long Software Tasks

Time Horizon Doubling Every ~7 Months

Source: Kwa, West, Becker et al. (METR) · arxiv:2503.14499 · March 2025

~7 Months

Doubling Time

$R^2 = 0.97$

Fit quality over 6 years

2 sec → ~2 hours

GPT-2 (2019) → o3 (2025)

A New Metric: 50% Task-Completion Time Horizon

A human-grounded measure of real-world AI autonomous capability, comparable across model generations.

01

Assemble Task Suite

170 diverse software & research tasks

- HCAST: 97 tasks (1 min–30 hrs)
- RE-Bench: 7 ML tasks (8 hrs each)
- SWAA: 66 micro-tasks (1–30 sec)

800+ human baselines
2,529 hours of human work logged

02

Measure Human Timing & AI Success Rates

For each task:

- Skilled humans timed on completion
- AI agents run on the same tasks
- Success/failure recorded per model

Logistic model fit to success rate vs. task duration for each AI model

03

Compute 50% Time Horizon

The 50% time horizon = task duration at which the AI succeeds 50% of the time

- Intuitive human-equivalent unit
- Comparable across all models
- Tracks capability growth over time

See methodology overview: [figure_2.jpg](#)

Family	Length	Description
find_shell_script	3 seconds	Multiple choice: “Which file is a shell script?” Choices: “run.sh”, “run.txt”, “run.py”, “run.md”
wikipedia_research	1 minute	Research simple factual information from Wikipedia and provide accurate answers to straightforward questions.
oxdna_simple	9 minutes	Detect and fix a bug in the input files for a molecular dynamics simulation using the oxDNA package.
munge_data	56 minutes	Write a Python script to transform JSON data from one format to another by inferring the conversion rules from provided example files.
cuda_backtesting	8 hours	Speed up a Python backtesting tool for trade executions by implementing custom CUDA kernels while preserving all functionality, aiming for a 30x performance improvement.

170 Tasks Spanning 3 Seconds to 8 Hours of Human Effort

HCAST • 97 tasks • 1 min – 30 hrs

RE-Bench • 7 tasks • 8 hours each

SWAA • 66 tasks • 1 sec – 30 sec

800+ human baselines • 2,529 hours of human work logged

Task Family	Suite	Human Time	Description
find_shell_script	SWAA	3 seconds	Multiple choice: "Which file is a shell script?"
wikipedia_research	SWAA	1 minute	Research simple factual information from Wikipedia
oxdna_simple	HCAST	9 minutes	Detect and fix a bug in molecular dynamics simulation input files
munge_data	HCAST	56 minutes	Write a Python script to transform JSON data by inferring conversion rules
cuda_backtesting	RE-Bench	8 hours	Implement custom CUDA kernels for 30x speedup of a backtesting tool

AI Task Horizon Grew from 2 Seconds (GPT-2) to ~1h 50m (o3) in 6 Years

~207 days

Doubling Time

$R^2 = 0.97$

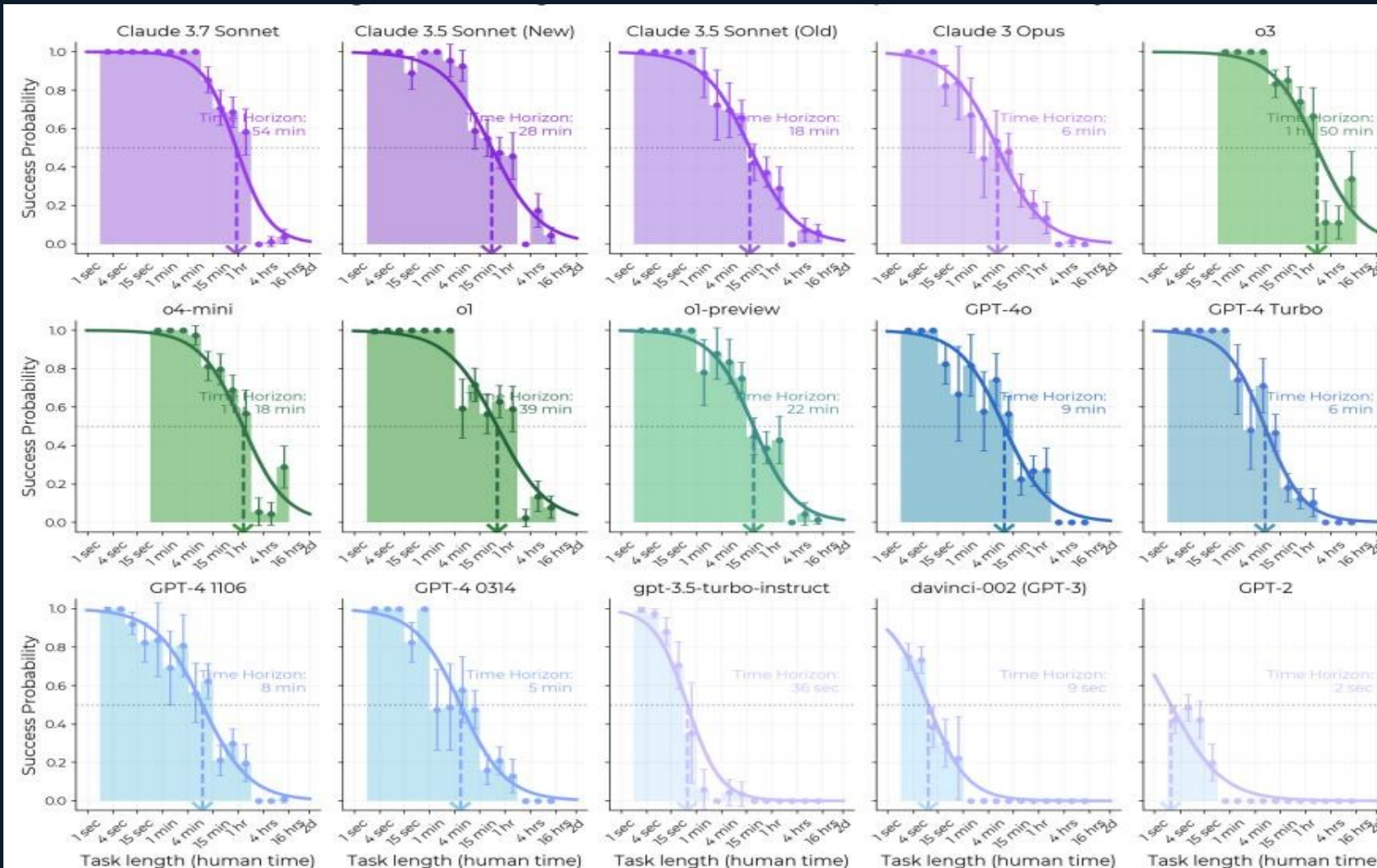
Model Fit Quality

6 Years

2019 → 2025

~55×

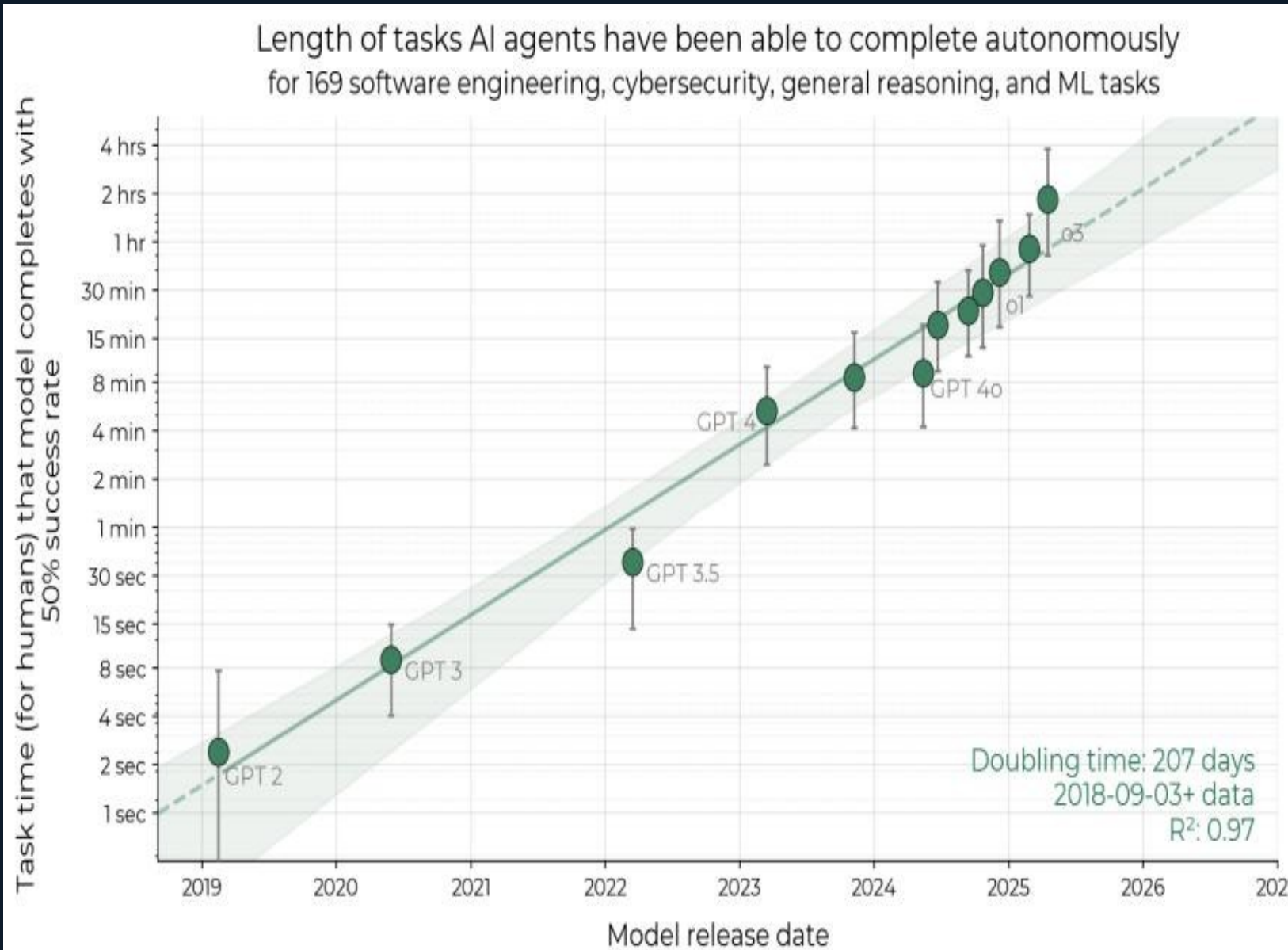
Growth Factor (sec → hr)



Key Milestones

- GPT-2 (2019) ~2 sec
- GPT-3 (davinci-002) (2020) ~9 sec
- GPT-3.5-turbo (2023) ~36 sec
- GPT-4 (0314) (2023) ~5 min
- GPT-4o (2024) ~9 min
- Claude 3.5 Sonnet (2024) ~28 min
- o1 (2024) ~39 min
- Claude 3.7 Sonnet (2025) ~54 min
- o3 (2025) ~1h 50m

Success Probability Drops Sharply as Task Length Increases for All Models



Reading the Curves

► Curve rightward shift

Each newer model's S-curve is pushed further right, reflecting higher 50% horizons.

► Frontier models

Claude 3.7 Sonnet (54 min) and o3 (~1h 50m) now succeed on tasks that GPT-4 (5 min) fails.

► Steep decline

All models show rapid drop-off in success rate as task duration grows —
no model maintains high accuracy beyond its time horizon.

► GPT-2 baseline

GPT-2 (2 sec) fails nearly all tasks longer than a few seconds;
the contrast against o3 shows ~55× growth.

► 80% horizon

80% success horizon is ~5× shorter than 50% horizon across models.

Reasoning, Tool Use, and Error Recovery Drive Time Horizon Growth

Four primary capability drivers have enabled frontier models to complete progressively longer autonomous tasks.

Improved Logical Reasoning

Models demonstrate stronger multi-step planning, causal inference, and abstract problem decomposition—enabling them to maintain coherent strategies across longer task horizons without losing context.

Better Tool Use

Enhanced ability to invoke shells, code interpreters, search APIs, and file systems reliably. Fewer tool-call failures and better handling of tool output errors translates directly into task completion gains.

Reliability & Self-Awareness

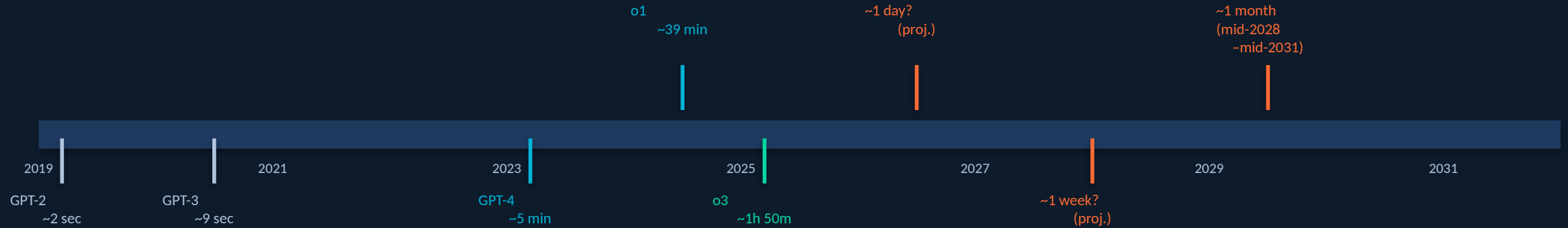
Frontier models better recognize when they are stuck or have made an error, reducing compounding mistakes. Greater calibration on uncertainty helps models avoid confident wrong actions.

Adaptation to Mistakes

The ability to detect and recover from intermediate errors without fully restarting—critical for long tasks where early mistakes would otherwise cascade into failure.

Extrapolation Predicts Month-Long Task Automation by 2028–2031

Projected 50% Time Horizon (Naive Extrapolation, ~207-day doubling)



Key Extrapolation Facts

- ~207 days doubling time ($R^2 = 0.97$ over 2019–2025)
- 2023–2025 growth rate is ~20% faster than overall trend
- 1-month horizon = 167 work hours ($\approx 10,020$ minutes)
- Naive extrapolation: 1-month horizon reached mid-2028 to mid-2031
- Possible acceleration since 2024 not yet baked into baseline

Caveats & Uncertainty

- Tasks may not represent real-world intellectual labor distributions
- Supplementary experiments found little evidence trends are slower on more realistic tasks
- Trend could accelerate or plateau — naive extrapolation only
- External validity of benchmark to actual job automation uncertain

Takeaways: Exponential Autonomous Capability Growth Demands Proactive Safety Governance

0 Robust, Intuitive Metric Validated at Scale

1 The 50% task-completion time horizon provides a human-grounded measure of AI autonomy with $R^2 = 0.97$ over 6 years of model releases across 170 tasks—making it a reliable benchmark for capability tracking.

0 Doubling Every ~7 Months — Possibly Accelerating

2 The ~207-day doubling time (2019–2025) appears to be accelerating: the 2023–2025 sub-period shows ~20% faster growth. Governance frameworks must anticipate continued rapid change.

0 Current Frontier: ~2 Hours of Human-Equivalent Work

3 o3 (2025) can autonomously complete tasks requiring ~1 hour 50 minutes of skilled human effort — up from 2 seconds for GPT-2 in 2019. This represents qualitatively new autonomous capability.

0 External Validity Remains an Open Question

4 The benchmark tasks may not fully represent real-world intellectual labor. Supplementary experiments found little evidence that trends are slower on realistic tasks, but the question remains open for safety policy.

0 Key Limitations: Unstructured Tasks & Generalization

5 Performance degrades significantly on less-structured, 'messy' tasks. AI safety and workforce automation assessments should account for this gap between controlled benchmarks and real-world deployment.